IGCSE 0607 Practice Questions — E1.1

Types of number

Total Questions: 9 — **Total Marks:** 56 Marks

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1 **(a)** Write 84 as a product of its prime factors.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 126.

..... **[2]**

2 Here is a list of numbers.

2 9 12 23 36 45 64 81

(a) Write down a number from the list which is:

(i) a multiple of 12,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) both a prime number and a factor of 46.

..... **[1]**

(b) Find the lowest common multiple (LCM) of 36 and 64.

..... **[2]**

3 **(a)** Let x be a rational number and y be an irrational number.
State whether $x + y$ is rational or irrational.

..... **[1]**

(b) Give an example of an irrational number y to show that y^2 can be a rational number.

[1]

(c) The nth term of a sequence is given by the expression

$$n^2 + n + 41$$

(i) Work out the 3rd term of this sequence, and state whether it is a prime number.

..... **[2]**

(ii) Explain why the 41st term of this sequence is not a prime number.

..... **[2]**

4 Let $m = p^3q$ and $n = pq^3$, where p and q are distinct prime numbers.

(a) Express, in terms of p and q:

(i) the highest common factor (HCF) of m and n,

..... **[1]**

(ii) the lowest common multiple (LCM) of m and n.

..... **[1]**

(b) The HCF of m and n is 15.
It is given that $m > n$.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(c) Show algebraically that the product $m \times n$ is a square number.

[2]

(d) Using your values of p and q, find the smallest positive integer k such that $m \times k$ is a cube number.

$k =$ ..... **[2]**

5 Three numbers A, B and C are given as products of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^x \times 5^2 \qquad C = 2^y \times 3 \times 7$$

The highest common factor (HCF) of A and B is 180.
The lowest common multiple (LCM) of B and C is 12600.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

6 Number A and Number B are given as products of their prime factors.

$A = 2^2 \times 3^3 \times 5$

$B = 2^x \times 3 \times 7$

(a) The highest common factor (HCF) of A and B is 6.
Find the value of x.

$x =$ ..... **[2]**

(b) Number $C = 2^2 \times 3^y \times 5^2$.
The lowest common multiple (LCM) of A and C is 8100.
Find the value of y.

$y =$ ..... **[3]**

(c) Here is a list of numbers.

$$\sqrt{11} \qquad -3.5 \qquad 4\pi \qquad \frac{7}{9} \qquad \sqrt{144} \qquad 0.\dot{4}$$

Write down all the irrational numbers from this list.

..... **[2]**

7 Here is a list of numbers.

2 9 15 16 23 27 64 72

(a) From the list, write down

(i) a prime number,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) a number that is both a square number and a cube number,

..... **[1]**

(iv) a triangle number.

..... **[1]**

(b) Express 72 as a product of its prime factors.

..... **[2]**

8 **(a)** Find the highest common factor (HCF) of 84 and 120.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 120.

..... **[2]**

(c) Write down the reciprocal of 0.8 as a fraction in its simplest form.

..... **[2]**

(d) State whether $\sqrt{120}$ is rational or irrational.
Give a reason for your answer.

..... **[2]**

9 **(a)** The number 168 can be written as $2^3 \times 3 \times 7$.
The number N can be written as $2^x \times 3^2 \times 7^y$, where x and y are integers.
The highest common factor (HCF) of 168 and N is 84.
The lowest common multiple (LCM) of 168 and N is 3528.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(b) Find the exact value of N.

$N =$ ..... **[1]**

(c) Two flashing lights, A and B, are turned on at the same time.
Light A flashes every 45 seconds.
Light B flashes every 72 seconds.
They both flash together at 10:15:00.
Work out the next time they flash together.

..... **[3]**